

MAT21-1

# Linear Algebra II

Determinants, eigenvalues, and diagonalization

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The intrinsic structure of a linear transformation, read from its stable directions.

Albert School — 34 hours

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## 1 Determinants

To each square matrix  $A \in M_n(\mathbb{R})$  is attached a scalar  $\det A$ , its **determinant**.

### 1.1 Computation

For a  $2 \times 2$  matrix,

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$$

For a  $3 \times 3$  matrix, **Sarrus' rule**:

$$\det \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33}.$$

In general the determinant is computed by **cofactor expansion**.

**Definition 1** Let  $A = (a_{ij}) \in M_n(\mathbb{R})$ . The **minor**  $M_{ij}$  is the determinant of the  $(n-1) \times (n-1)$  matrix obtained by deleting row  $i$  and column  $j$  of  $A$ , and the **cofactor** is  $C_{ij} = (-1)^{i+j}M_{ij}$ . Expansion along row  $i$  gives

$$\det A = \sum_{j=1}^n a_{ij}C_{ij},$$

and the analogous formula holds for expansion along any column.

The determinant may also be computed by **row reduction**, using the effect of the elementary row operations stated in the properties below: reduce  $A$  to a triangular matrix, for which the determinant is the product of the diagonal entries.

## 1.2 Geometric interpretation

The determinant is the **signed volume scaling factor** of the linear map  $X \mapsto AX$ : the image of the unit  $n$ -cube is a parallelepiped of volume  $|\det A|$ , and the sign of  $\det A$  records whether the map preserves (+) or reverses (−) orientation.

## 1.3 Properties

**Proposition 2** For  $A, B \in M_n(\mathbb{R})$ :

- **Multiplicativity:**  $\det(AB) = \det(A) \det(B)$ .
- **Transpose invariance:**  $\det(A^T) = \det(A)$ .
- **Invertibility criterion:**  $A$  is invertible if and only if  $\det A \neq 0$ , and then  $\det(A^{-1}) = 1/\det(A)$ .

**Proposition 3** Effect of the elementary row operations on  $\det A$ :

- swapping two rows multiplies the determinant by  $-1$ ;
- multiplying a row by  $\lambda$  multiplies the determinant by  $\lambda$ ;
- adding a multiple of one row to another leaves the determinant unchanged.

In particular  $\det A = 0$  whenever two rows are equal or a row is zero.

## 2 Eigenvalues and eigenvectors

**Definition 4** Let  $A \in M_n(\mathbb{R})$ . A scalar  $\lambda \in \mathbb{R}$  is an **eigenvalue** of  $A$  if there is a non-zero vector  $v$ , called an **eigenvector**, with

$$Av = \lambda v.$$

The set

$$E_\lambda = \ker(A - \lambda I) = \{v : Av = \lambda v\}$$

is the **eigenspace** associated with  $\lambda$ ; it is a subspace, and its non-zero elements are the eigenvectors for  $\lambda$ .

### 2.1 Characteristic polynomial

A scalar  $\lambda$  is an eigenvalue exactly when  $A - \lambda I$  is not invertible, i.e. when  $\det(A - \lambda I) = 0$ .

**Definition 5** The **characteristic polynomial** of  $A \in M_n(\mathbb{R})$  is

$$\chi_A(\lambda) = \det(A - \lambda I),$$

a polynomial of degree  $n$  in  $\lambda$ . Its roots are exactly the eigenvalues of  $A$ .

## 2.2 Trace and determinant

**Proposition 6** Counting each eigenvalue with its multiplicity as a root of  $\chi_A$ :

- the **trace** of  $A$  (sum of diagonal entries) equals the sum of the eigenvalues;
- the determinant of  $A$  equals the product of the eigenvalues.

## 3 Algebraic and geometric multiplicity

**Definition 7** Let  $\lambda$  be an eigenvalue of  $A$ .

- Its **algebraic multiplicity** is its multiplicity as a root of the characteristic polynomial  $\chi_A$ .
- Its **geometric multiplicity** is  $\dim E_\lambda$ , the dimension of the associated eigenspace.

**Proposition 8** For every eigenvalue,  $1 \leq \text{geometric multiplicity} \leq \text{algebraic multiplicity}$ .

The distinction matters because diagonalizability requires equality for every eigenvalue, as stated next.

## 4 Diagonalization

**Definition 9** A matrix  $A \in M_n(\mathbb{R})$  is **diagonalizable** if there exist an invertible matrix  $P$  and a diagonal matrix  $D$  with

$$A = PDP^{-1}.$$

When this holds, the diagonal entries of  $D$  are the eigenvalues of  $A$  and the columns of  $P$  are corresponding eigenvectors (in matching order).

**Theorem 10 (Diagonalization criterion)**  $A \in M_n(\mathbb{R})$  is diagonalizable over  $\mathbb{R}$  if and only if both:

1. the characteristic polynomial **splits** over  $\mathbb{R}$  (it is a product of degree-one factors, i.e. all roots are real); and
2. for every eigenvalue, the geometric multiplicity equals the algebraic multiplicity.

### 4.1 Procedure

To diagonalize  $A$ :

1. compute  $\chi_A$  and its roots, the eigenvalues;
2. for each eigenvalue  $\lambda$ , solve  $(A - \lambda I)v = 0$  to obtain a basis of  $E_\lambda$ ;
3. if the eigenvectors collected from all eigenspaces number  $n$ , place them as the columns of  $P$  and the corresponding eigenvalues on the diagonal of  $D$ ; then  $A = PDP^{-1}$ .

### 4.2 Matrix powers

**Proposition 11** If  $A = PDP^{-1}$  with  $D = \text{diag}(\lambda_1, \dots, \lambda_n)$ , then for every integer  $n \geq 0$ ,

$$A^n = PD^nP^{-1}, \quad D^n = \text{diag}(\lambda_1^n, \dots, \lambda_n^n).$$

This reduces computing a power of  $A$  to taking powers of scalars.

### 4.3 Linear recurrence relations

A vector recurrence  $u_{n+1} = Au_n$  has solution  $u_n = A^n u_0$ . When  $A$  is diagonalizable,  $A^n = PD^nP^{-1}$  gives a closed form for  $u_n$ , and hence for each scalar sequence appearing in it.

## 5 The Cayley–Hamilton theorem

**Theorem 12 (Cayley–Hamilton)** Every square matrix satisfies its own characteristic polynomial:

$$\chi_A(A) = 0,$$

where the constant term  $c_0$  of  $\chi_A$  is read as  $c_0I$ .

**Verification** for a given  $A$  consists of forming  $\chi_A$ , substituting  $A$  for the variable, and checking that the resulting matrix is the zero matrix.

### 5.1 Inverse as a polynomial in $A$

Write  $\chi_A(\lambda) = (-1)^n(\lambda^n + c_{n-1}\lambda^{n-1} + \dots + c_1\lambda + c_0)$ . If  $A$  is invertible then  $c_0 = (-1)^n \det A \neq 0$ , and Cayley–Hamilton gives

$$A^{-1} = -\frac{1}{c_0}(A^{n-1} + c_{n-1}A^{n-2} + \dots + c_1I),$$

expressing  $A^{-1}$  as a polynomial in  $A$ .

### 5.2 Powers of a non-diagonalizable matrix

Cayley–Hamilton shows  $A^n$  equals a polynomial in  $A$  of degree at most  $n-1$ : reduce any higher power modulo  $\chi_A$ . This computes powers of  $A$  even when  $A$  is not diagonalizable.

## 6 Matrix inverse

### 6.1 Via Gaussian elimination

Form the augmented matrix  $(A \mid I)$  and apply Gaussian elimination to reduce the left block to  $I$ . If this succeeds, the right block becomes  $A^{-1}$ :

$$(A \mid I) \sim \dots \sim (I \mid A^{-1}).$$

If the left block cannot be reduced to  $I$  (a zero pivot appears),  $A$  is not invertible.

### 6.2 Via Cayley–Hamilton

Alternatively, when  $\det A \neq 0$ , compute  $A^{-1}$  as the polynomial in  $A$  given above.

## 7 Applications

### 7.1 Graph adjacency matrices

The **adjacency matrix**  $A$  of a graph on  $n$  vertices has  $a_{ij}$  equal to the number of edges from vertex  $i$  to vertex  $j$ .

**Proposition 13** For every integer  $k \geq 1$ , the entry  $(A^k)_{ij}$  is the number of **walks of length  $k$**  from vertex  $i$  to vertex  $j$ .

The long-run behaviour of a linear iteration on the graph is governed by the **dominant eigenvalue** (largest in absolute value): its eigenvector captures the asymptotic direction along which the iteration grows.

### 7.2 Markov chains via stochastic matrices

**Definition 14** A square matrix  $P$  with non-negative entries is **column-stochastic** if each column sums to 1 (**row-stochastic** if each row sums to 1). A probability distribution is a non-negative vector whose entries sum to 1.

For a column-stochastic  $P$ , a distribution evolves by  $x_{k+1} = Px_k$ .

**Proposition 15** A column-stochastic matrix always has 1 as an eigenvalue. A **stationary distribution** is a probability distribution  $\pi$  with

$$P\pi = \pi,$$

i.e. an eigenvector for the eigenvalue 1.

When  $P$  is diagonalizable and 1 is its dominant eigenvalue (all other eigenvalues having absolute value  $< 1$ ), writing  $P = QDQ^{-1}$  and computing  $x_k = P^k x_0 = QD^k Q^{-1} x_0$  shows that  $x_k$  converges, from any initial distribution  $x_0$ , to the stationary distribution  $\pi$ .

## 8 Similar matrices and change of basis

**Definition 16** Two matrices  $A, B \in M_n(\mathbb{R})$  are **similar** if there is an invertible  $P$  with  $B = P^{-1}AP$ .

Similar matrices have the same characteristic polynomial, hence the same eigenvalues, determinant, and trace.

By the change-of-basis formula (MAT11-3), if  $A$  is the matrix of a linear map in one basis and  $P$  is the change-of-basis matrix to a second basis, then  $P^{-1}AP$  is the matrix of the **same** map in the second basis. Similar matrices therefore represent one linear transformation in different coordinate systems, and diagonalization is the choice of a basis of eigenvectors in which that transformation acts by independent scalings along each axis.